

$$1. h\nu = \frac{1}{2}mv_m^2 + A$$

(1) 光电效应 "红限"现象
截止频率

$$(2) \text{黑体辐射 Planck公式 } M(\lambda, T) / M(\nu, T) \rightarrow \text{Stefan公式 } M(T) = \sigma T^4$$

$$M(T) = \int_0^\infty M(\lambda, T) d\lambda / \int_0^\infty M(\nu, T) d\nu$$

Stefan公式 测温仪器: 红外线测温仪

(3) Bohr量子理论 { 成功: 能级 解释一价碱金属的电子光谱
不成功: 经典理论+量子条件

(4) 固体与射的比热问题 $\rightarrow$ 量子概念: 自旋理论, 电子自旋

$$2. \text{氢原子 } \psi_{100} = \frac{1}{4} R_{21}(r) Y_{11}(0, \varphi) - \frac{\sqrt{2}}{4} R_{21}(r) Y_{10}(0, \varphi) + \frac{1}{\sqrt{2}} R_{21}(r) Y_{1-1}(0, \varphi)$$

(1) E 可取值 $0 \frac{\pi}{2}$

(2) L^2 可取值 $0 \frac{\pi}{2}$

(3) L_z 可取值 $0 \frac{\pi}{2}$

$$\psi_{100} = \frac{1}{4} R_{21}(r) Y_{11}(0, \varphi) - \frac{\sqrt{2}}{4} R_{21}(r) Y_{10}(0, \varphi) + \frac{1}{\sqrt{2}} R_{21}(r) Y_{1-1}(0, \varphi)$$

$$\sum |C_n|^2 = (\frac{1}{4})^2 + (\frac{\sqrt{2}}{4})^2 + (\frac{1}{\sqrt{2}})^2 = 1 \quad \text{归一}$$

$$(1) \text{氢原子 } E_n = -\frac{me_0^4}{2\hbar^2 n^2}$$

$$n=2 \quad E_2 = -\frac{me_0^4}{8\hbar^2} \quad P_2 = \frac{1}{16} + \frac{7}{16} = \frac{1}{2}$$

$$n=3 \quad E_3 = -\frac{me_0^4}{18\hbar^2} \quad P_3 = \frac{1}{2}$$

$$(2) L^2 = l(l+1)\hbar^2$$

$$l=1 \quad L^2 = 2\hbar^2 \quad P = 1 \quad \text{确定值}$$

$$(3) L_z = m\hbar$$

$$m=1 \quad L_z = \hbar \quad P_1 = \frac{1}{16}$$

$$m=0 \quad L_z = 0 \quad P_0 = \frac{7}{16}$$

$$m=-1 \quad L_z = -\hbar \quad P_{-1} = \frac{1}{16}$$

$$3. \quad m \quad V(x)$$

$$(1) V(x) = \begin{cases} \infty & |x| > a \\ 0 & |x| \leq a \end{cases} \rightarrow E$$

$$\text{一维无限深方势阱: } E_n = \frac{n^2 \pi^2 \hbar^2}{8ma^2} \quad n=1, 2, 3, \dots$$

$$(2) V(x) = a\delta(x) \quad a < 0 \rightarrow 0 E \quad \text{特征长度}$$

$$E < 0 \quad \text{令 } E = -|E|$$

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + a\delta(x)\psi = E\psi$$

$$x \neq 0: -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi$$

$$\text{令 } k = \sqrt{\frac{2m|E|}{\hbar^2}}$$

$$\frac{d^2\psi}{dx^2} = k^2\psi$$

$$\psi_1(x) = Ae^{kx} \quad x < 0$$

$$\psi_2(x) = Be^{-kx} \quad x > 0$$

$$\begin{cases} \psi_1(0) = \psi_2(0): A = B \\ \psi_1'(0) - \psi_2'(0) = \frac{2ma}{\hbar^2} \psi_1(0): -Ak - Ak = \frac{2ma}{\hbar^2} A \end{cases}$$

$$k = -\frac{ma}{\hbar^2}$$

$$E = -\frac{\hbar^2 k^2}{2m} = -\frac{ma^2}{2\hbar^2}$$

$$\text{特征长度 } \lambda_0 = \frac{1}{|k|}$$

$$X_0 = |k| = \frac{\hbar^2}{m|a|}$$

$$13) \quad V(x) = \begin{cases} V_0 \delta(x) & |x| < a \\ \infty & |x| \geq a \end{cases} \quad V_0 > 0 \rightarrow E \text{ 满足的方程}$$

$$E > 0$$

$$|x| \geq a: \psi(x) = 0$$

$$|x| < a: -\frac{\hbar^2}{2m} \frac{d^2 \psi(x)}{dx^2} + V_0 \delta(x) \psi(x) = E \psi(x)$$

$$x \neq 0: -\frac{\hbar^2}{2m} \frac{d^2 \psi(x)}{dx^2} = E \psi(x)$$

$$\text{令 } k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\frac{d^2 \psi(x)}{dx^2} = -k^2 \psi(x)$$

$$\begin{cases} \psi(0^+) = \psi(0^-) \\ \psi'(0^+) - \psi'(0^-) = \frac{2mV_0}{\hbar^2} \psi(0) \end{cases}$$

$$\psi(-x) = \psi(x) \rightarrow \text{确定的奇偶性}$$

$$\textcircled{1} \text{ 奇对称 } \psi(-x) = -\psi(x) \quad \psi(0) = 0$$

$$\psi(0^+) = \psi(0^-)$$

$$-\text{势无限深方势阱:}$$

$$\psi(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi}{2a} (x+a) \quad n = 2, 4, 6, \dots$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{8ma^2}$$

$$\textcircled{2} \text{ 偶对称 } \psi(-x) = \psi(x)$$

$$\psi_1(x) = A \cos kx + B \sin kx \quad 0 < x < a$$

$$\psi_2(x) = A \cos kx - B \sin kx \quad -a < x < 0$$

$$\begin{cases} \psi_1(a) = 0: A \cos ka + B \sin ka = 0 \\ \psi_1(0^+) - \psi_2(0^-) = \frac{2mV_0}{\hbar^2} \psi_1(0): B = \frac{2mV_0}{\hbar^2} A \end{cases}$$

$$\tan ka = -\frac{A}{B} = -\frac{\hbar^2}{2mV_0} = -\frac{\hbar^2}{mV_0} ka \quad b = \frac{\hbar^2}{mV_0} \text{ 势垒的特征长度}$$

$$14)$$

4.

11) 厄米算符证明! $\hat{A} = i\hbar x \frac{\partial}{\partial x}$

$$\hat{A}^\dagger = \hat{A}$$

$$\hat{A}^\dagger = (i\hbar x \frac{\partial}{\partial x})^\dagger = -i\hbar \frac{\partial}{\partial x} x \neq \hat{A}$$

$$(-x \hat{p})^\dagger = (\hat{p} x)$$

$\hat{A}$ 不是

12) $[\hat{x}, \hat{p}] = i\hbar \rightarrow [\hat{x}^2, \hat{p}^2]$

$$[\hat{x}^2, \hat{p}^2] = \hat{x} [\hat{x}, \hat{p}^2] + [\hat{x}, \hat{p}^2] \hat{x}$$

$$= 2i\hbar \hat{x} \hat{p} + 2i\hbar \hat{p}^2 \hat{x}$$

$$= 2i\hbar (\hat{x} \hat{p}^2 + \hat{p}^2 \hat{x})$$

13) 证明正交

$$\begin{cases} \hat{F} \psi_n = F_n \psi_n \\ \hat{F} \psi_m = F_m \psi_m \end{cases}$$

$$\langle \psi_m, \hat{F} \psi_n \rangle = F_n \langle \psi_m, \psi_n \rangle = \langle \hat{F} \psi_m, \psi_n \rangle = F_m \langle \psi_m, \psi_n \rangle$$

$$(F_m - F_n) \langle \psi_m, \psi_n \rangle = 0$$

$$F_m \neq F_n \rightarrow \langle \psi_m, \psi_n \rangle = 0$$

14) 厄米算符

平均值实数

5. 一维谐振子 m $V(x) = \frac{1}{2} m (\omega_1^2 x^2 + \omega_2 x + c)$

$$\langle \hat{x} \rangle_{t=0} = 0 \quad \langle \hat{p} \rangle_{t=0} = 0$$

$\rightarrow \langle \hat{x} \rangle$ 与 t 的关系

$$\frac{d\langle \hat{x} \rangle}{dt} = \langle \frac{\partial \hat{x}}{\partial t} \rangle + \frac{1}{i\hbar} [\hat{x}, \hat{H}]$$

$$\frac{d\langle \hat{x} \rangle}{dt} = \frac{\partial \hat{x}}{\partial t} + \frac{1}{i\hbar} [\hat{x}, \hat{H}]$$

$$= 0 + \frac{1}{i\hbar} [\hat{x}, \frac{\hat{p}^2}{2m}]$$

$$= \frac{1}{i\hbar} \frac{1}{2m} 2i\hbar \langle \hat{p} \rangle$$

$$= \frac{\langle \hat{p} \rangle}{m}$$

$$\frac{d\langle \hat{p} \rangle}{dt} = \langle \frac{\partial \hat{p}}{\partial t} \rangle + \frac{1}{i\hbar} [\hat{p}, \hat{H}]$$

$$= 0 + \frac{1}{i\hbar} [\hat{p}, \frac{1}{2} m (\omega_1^2 \hat{x}^2 + \omega_2 \hat{x} + c)]$$

$$= \frac{1}{i\hbar} \frac{m}{2} (\omega_1^2 (i\hbar \hat{x}) + \omega_2 (i\hbar))$$

$$= -\frac{m}{2} (2\omega_1^2 \langle \hat{x} \rangle + \omega_2)$$

$$\left\{ \begin{aligned} \frac{d\langle \hat{x} \rangle}{dt} &= \frac{1}{m} \frac{d\langle \hat{p} \rangle}{dt} \\ &= -\omega_1^2 \langle \hat{x} \rangle + \frac{1}{2} \omega_2 \end{aligned} \right.$$

$$\frac{d\langle \hat{p} \rangle}{dt} = -m\omega_1^2 \langle \hat{x} \rangle - \frac{1}{2} m \omega_2$$

$$\langle \hat{p} \rangle = A \sin \omega_1 t \quad |\langle \hat{p} \rangle_{t=0} = 0|$$

$$\frac{d\langle \hat{x} \rangle}{dt} = \frac{\langle \hat{p} \rangle}{m} = \frac{A \sin \omega_1 t}{m}$$

$$\langle \hat{x} \rangle = -\frac{A}{m\omega_1} \cos \omega_1 t + C$$

$$\text{由 } \langle \hat{x} \rangle_{t=0} = 0 : C = \frac{A}{m\omega_1}$$

$$\langle \hat{x} \rangle = -\frac{A}{m\omega_1} (\cos \omega_1 t - 1)$$

$$\text{由 } \frac{d\langle \hat{p} \rangle}{dt} = -m\omega_1^2 \langle \hat{x} \rangle - \frac{1}{2} m \omega_2$$

$$\text{or } \frac{d\langle \hat{x} \rangle}{dt} = -\omega_1^2 \langle \hat{x} \rangle + \frac{1}{2} \omega_2$$

$$\rightarrow A = -\frac{m\omega_2}{2\omega_1}$$

$$\langle \hat{x} \rangle = \frac{\omega_2}{2\omega_1^2} (\cos \omega_1 t - 1)$$

b. 117 $|2\rangle \rightarrow E_1, E_2$

$$\langle 1 | \hat{H} | 1 \rangle = E_1 + b \quad \langle 2 | \hat{H} | 2 \rangle = E_2 + b \quad \langle 1 | \hat{H} | 2 \rangle = \langle 2 | \hat{H} | 1 \rangle = a$$

→ ① - 级近似量

② - 二级近似量

$$\textcircled{1} \quad \hat{H} = \begin{pmatrix} E_1+b & a \\ a & E_2+b \end{pmatrix} = \begin{pmatrix} E_1 & 0 \\ 0 & E_2 \end{pmatrix} + \begin{pmatrix} b & a \\ a & b \end{pmatrix}$$

$$E_n^{(1)} = \langle \psi_n^{(0)} | \hat{H}' | \psi_n^{(0)} \rangle = H_{nn}$$

$$|1\rangle: E_1 = E_1^{(0)} + E_1^{(1)} = E_1 + b$$

$$|2\rangle: E_2 = E_2^{(0)} + E_2^{(1)} = E_2 + b$$

$$\textcircled{2} \quad E_n^{(2)} = \sum_{m \neq n} \frac{\langle \psi_m^{(0)} | \hat{H}' | \psi_n^{(0)} \rangle \langle \psi_n^{(0)} | \hat{H}' | \psi_m^{(0)} \rangle}{E_n^{(0)} - E_m^{(0)}} = \sum_{m \neq n} \frac{|H_{mn}|^2}{E_n^{(0)} - E_m^{(0)}}$$

$$E_1^{(2)} = \frac{|H_{21}|^2}{E_1^{(0)} - E_2^{(0)}} = \frac{a^2}{E_1 - E_2}$$

$$E_2^{(2)} = \frac{|H_{12}|^2}{E_2^{(0)} - E_1^{(0)}} = \frac{a^2}{E_2 - E_1}$$

$$|1\rangle: E_1 = E_1^{(0)} + E_1^{(1)} + E_1^{(2)} = E_1 + b + \frac{a^2}{E_1 - E_2}$$

$$|2\rangle: E_2 = E_2^{(0)} + E_2^{(1)} + E_2^{(2)} = E_2 + b + \frac{a^2}{E_2 - E_1}$$

7. m 散射 有效质量 μ

8. 证明固体物理中常用的TKK等式:

$$\sum_n |E_n - E_a| |\langle n | X | a \rangle|^2 = \frac{\hbar^2}{2m} \quad \text{其中 } |n\rangle |a\rangle \text{ 为基态的 } T \text{ 位置算子}$$

E_n, E_a 为位置 T 算子的能量

$$\sum_n |E_n - E_a| |\langle n | X | a \rangle|^2$$

$$= \sum_n |E_n - E_a| \langle a | X | n \rangle \langle n | X | a \rangle \quad \text{对应!}$$

$$= \frac{1}{2} \sum_n [\underbrace{|E_n - E_a| \langle a | X | n \rangle \langle n | X | a \rangle}_{\text{对应!}} - \underbrace{|E_n - E_a| \langle a | X | n \rangle \langle n | X | a \rangle}_{\text{对应!}}]$$

$$[\hat{X}, \hat{H}] = [\hat{X}, \frac{\hat{p}^2}{2m} + V(x)] = \frac{i\hbar}{m} \hat{p}$$

$$\hat{p} = \frac{m}{i\hbar} [\hat{X}, \hat{H}]$$

$$\langle n | \hat{p} | a \rangle = \frac{m}{i\hbar} \langle n | \hat{X} \hat{H} - \hat{H} \hat{X} | a \rangle = \frac{m}{i\hbar} (E_n - E_a) \langle n | X | a \rangle$$

$$\langle a | \hat{p} | n \rangle = \frac{m}{i\hbar} \langle a | \hat{X} \hat{H} - \hat{H} \hat{X} | n \rangle = \frac{m}{i\hbar} (E_n - E_a) \langle a | X | n \rangle$$

$$\sum_n |E_n - E_a| |\langle n | X | a \rangle|^2$$

$$= \frac{1}{2} \sum_n [\frac{i\hbar}{m} \langle a | \hat{p} | n \rangle \langle n | \hat{X} | a \rangle - \frac{i\hbar}{m} \langle a | \hat{X} | n \rangle \langle n | \hat{p} | a \rangle]$$

$$= \frac{i\hbar}{2m} [\langle a | \hat{p} \hat{X} | n \rangle \langle n | \hat{X} | a \rangle - \langle a | \hat{X} \hat{p} | n \rangle \langle n | \hat{X} | a \rangle]$$

$$= \frac{i\hbar}{2m} \langle a | \hat{p} \hat{X} - \hat{X} \hat{p} | a \rangle$$

$$= \frac{i\hbar}{2m} \langle a | [\hat{p}, \hat{X}] | a \rangle$$

$$= \frac{\hbar^2}{2m} \quad -i\hbar$$